

Statistics for Data Science & Analysis

A Unified Curriculum and Study Roadmap for Analysts and Scientists

This comprehensive study guide bridges foundational statistical theory with advanced, real-world industry techniques. Designed for both Data Analysts and Data Scientists, it structures core topics—ranging from central tendency up to causal inference—into actionable modules. Use this guide to master descriptive baseline analytics, probabilistic distributions, experimentation, and predictive statistical modeling.

Module 1: Descriptive Foundations & Outlier Dynamics

Descriptive statistics form the bedrock of data exploration, allowing us to summarize and understand the shape of our data. However, choosing the right measure of central tendency requires a deep understanding of outlier dynamics.

Measures of Central Tendency

- **Mean:** The mathematical average (sum divided by count). Highly sensitive to extreme values, making it volatile for skewed distributions.
- **Median:** The positional middle value of a sorted dataset. Highly robust because it ignores numerical magnitudes, focusing strictly on order.
- **Mode:** The most frequent data value in a distribution. While valuable, it has specific advantages and distinct structural risks.

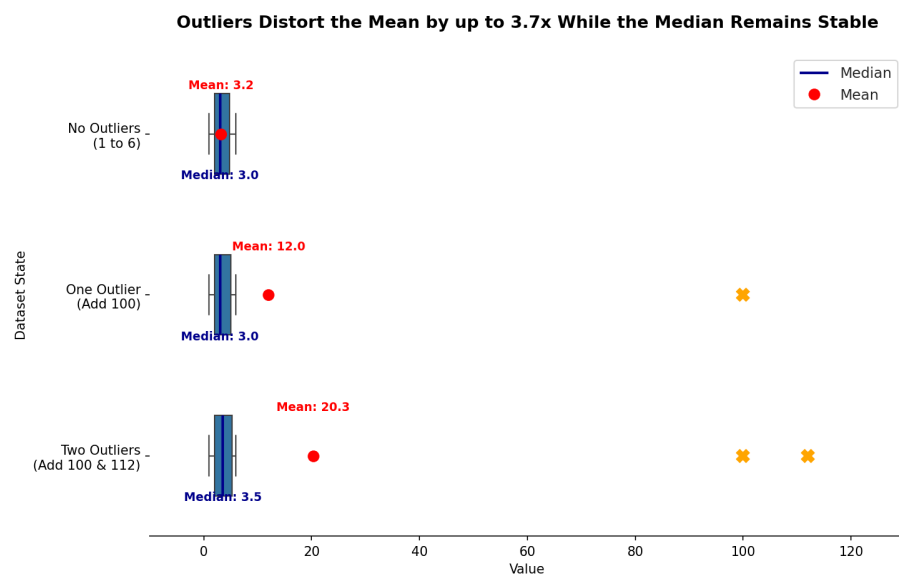


Figure: The volatility of the Mean versus the robustness of the Median under extreme outliers (100 and 112).

The Mode: Applications & Limitations

In data science, the **Mode** is extremely powerful for handling missing categorical (qualitative) variables via **Mode Imputation** (e.g., filling in missing flower categories like Rose, Lily, or Sunflower with the most frequent class, Lily, as shown in *categorical-mode-imputation.png*). However, a critical limitation occurs with repeated outliers. If multiple identical outliers are introduced (e.g., adding several 100s), they will hijack the frequency counts, shifting the mode completely away from the true central distribution (visualized in *mode-shift-outlier.png*).

Module 1 (Cont.): Dispersion, Variance, and Z-Scores

Central tendency only tells half the story. Two datasets can share the exact same mean of 2.0 while having completely different spreads, as visualized below. To quantify this spread, we must utilize measures of dispersion.

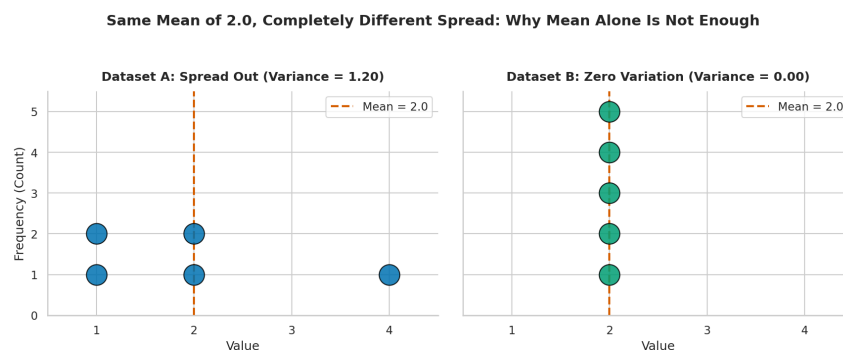


Figure: Why mean is not enough: Dataset A (dispersed) vs. Dataset B (fully centered on 2.0).

Variance: Quantifying the Spread

Variance measures the average squared distance of each data point from the mean. We square the distances to prevent positive and negative differences from canceling each other out, and to exponentially penalize larger deviations from the center. When working with samples, we use **Bessel's Correction** (dividing by $n - 1$ instead of n) to correct for the inherent bias in estimating population dispersion from small samples.

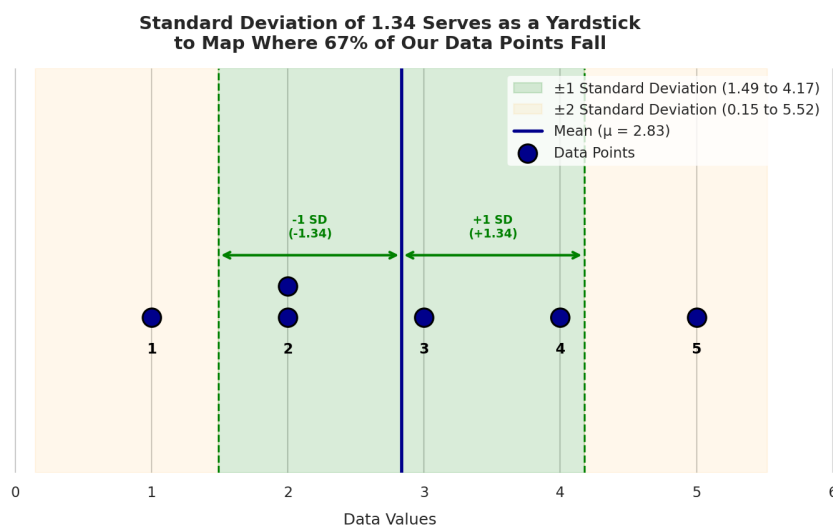


Figure: Standard Deviation as a standardized spatial yardstick (1.34) around the central mean (2.83).

Standard Deviation & The Z-Score Yardstick

By taking the square root of variance, **Standard Deviation** returns the dispersion metric to the original data's unit, serving as a standard statistical yardstick. From standard deviation, we derive the **Z-score**, which measures how many standard deviations a specific data point lies from the mean. Applying a Z-score transformation to all numerical columns re-scales them to a mean of 0 and standard deviation of 1. This process, called **Standardisation**, is a critical preprocessing step for machine learning models (e.g., KNN, SVM, Neural Networks) to prevent variables with larger scales from dominating the loss function.

Module 2: Probability & Distribution Blueprints

A distribution mathematically defines 'how your data looks' and is modeled by a **Probability Density Function (PDF)** for continuous variables or a **Probability Mass Function (PMF)** for discrete categorical variables. Understanding these standard blueprints is essential for modeling real-world phenomena.

Core Statistical Distributions

- **Gaussian (Normal) Distribution:** The classic, symmetrical bell curve where mean = median = mode. Guided by the **Empirical Rule (68-95-99.7)**: 68% of data falls within ± 1 SD, 95% within ± 2 SD, and 99.7% within ± 3 SD.
- **Standard Normal Distribution:** A standardized Gaussian curve with mean = 0 and standard deviation = 1. Used in hypothesis testing and feature scaling.
- **Log-Normal Distribution:** A highly right-skewed distribution where the log of the variable is normally distributed. Frequently models wealth distributions, housing prices, or user comment lengths.
- **Bernoulli & Binomial:** Bernoulli models a single trial with binary outcomes (success/failure). Binomial models the number of successes across successive independent Bernoulli trials.
- **Pareto Distribution (Power Law):** The 80/20 rule (e.g., 80% of business sales come from 20% of products). Highly unequal, non-Gaussian distribution.

The Central Limit Theorem (CLT)

The CLT is the ultimate bridge of statistics. It states that regardless of the shape of your original population distribution (skewed, uniform, or random), the distribution of the **sample means** will always approach a normal distribution as the sample size increases (typically $n > 30$). This mathematical guarantee is what allows analysts to perform parametric hypothesis tests on highly skewed real-world data.

Class Imbalance Trap (For Scientists): For rare events like fraud detection, ad click conversions, or medical diagnoses (e.g., 1% positive class), a model that predicts 'no fraud' for 100% of cases achieves 99% accuracy but is completely useless. In these scenarios, never rely on standard **Accuracy**. Instead, evaluate models using **Precision** (minimising false positives), **Recall** (minimising false negatives), and the **F1-Score** (the harmonic mean of both) to gauge true performance.

Module 3: Inferential Testing & Hypothesis Foundations

Inferential statistics allows us to make confident claims about an entire population using only a representative sample. This requires setting up formal hypothesis tests and understanding parametric vs. non-parametric boundaries.

Hypothesis Testing Framework

- **Null Hypothesis (H_0):** Assumes no effect, change, or difference exists. We aim to reject this based on evidence.
- **Alternative Hypothesis (H_1):** The claim we support if H_0 is successfully rejected.
- **Alpha Level (α):** The threshold for significance (commonly 0.05). It is the probability of committing a Type I error (false positive).
- **P-Value:** The probability of obtaining results at least as extreme as the observed data, assuming H_0 is true. If $p < \alpha$, we reject the null hypothesis.

Parametric vs. Non-Parametric Boundaries

Parametric tests (like the **Z-test** and **T-test**) assume the underlying data follows a normal distribution. The Z-test is used when the population standard deviation is known and sample size is large ($n > 30$); the T-test is used when the population standard deviation is unknown or the sample size is small.

When real-world data is highly skewed or ordinal (failing normality assumptions), analysts must use **Non-Parametric Tests** to maintain mathematical validity:

- **Mann-Whitney U Test:** Compares differences between two independent groups without assuming normality (non-parametric alternative to the independent two-sample T-test).
- **Kruskal-Wallis Test:** Compares differences among three or more groups (non-parametric alternative to a one-way ANOVA).
- **Chi-Square Test of Independence:** Determines whether there is a statistically significant association between two categorical variables.

Modules 4 & 5: A/B Testing & Advanced Data Science Tracks

While general hypothesis testing is academic, business intelligence requires rigorous experimentation and predictive modeling tailored to specific roles.

Module 4: Business Experimentation (Data Analyst Track)

- **A/B Testing Experimental Design:** Setting up randomized controlled experiments to test product changes. Always conduct a **Power Analysis** beforehand to estimate the minimum required sample size based on expected effect size, statistical power ($1 - \beta$), and alpha level.
- **Multiple Comparison Problem:** Testing multiple variations (e.g., A vs. B vs. C) spikes the probability of a false positive. Analysts must use the **Bonferroni Correction** (dividing alpha by the number of comparisons) to control this risk.
- **Statistical vs. Practical Significance:** Large samples (e.g., millions of users) can make tiny, commercially meaningless changes statistically significant. Analysts must always evaluate if the practical business return justifies implementation costs.

Simpson's Paradox Trap (For Analysts): Occurs when a statistical trend appears in several different groups of data but completely disappears or reverses when these groups are aggregated. For example, a marketing campaign might show a higher conversion rate for Version B overall, but when segmented by region, Version A actually wins in every single individual territory. Always segment and drill down before drawing final business conclusions!

Module 5: Advanced Statistical Modeling (Data Scientist Track)

- **Frequentist vs. Bayesian Statistics:** Frequentists treat population parameters as fixed, unknown constants. Bayesians treat them as random variables, incorporating **Prior Probabilities** and updating them into **Posterior Probabilities** as data is collected. Tech leaders (Netflix, Airbnb) use Bayesian A/B testing to directly ask: *'What is the probability that Version B is better than A?'*
- **Causal Inference:** Establishing causality when randomized A/B tests are impossible due to ethical or operational limits. Methods include **Propensity Score Matching** (creating artificial control groups) and **Difference-in-Differences** (analyzing temporal shifts before and after a policy change).
- **Time Series Autocorrelation:** Recognizing that chronological data points are dependent on their own historical values, violating the standard independence assumptions of linear modeling.

Statistical Applications: Focus & Roadmap

This section outlines how Data Analysts and Data Scientists focus on different aspects of statistics, followed by an actionable 8-week learning roadmap.

Data Analyst vs. Data Scientist Focus

Statistical Domain	Data Analyst Focus	Data Scientist Focus
Central Tendency & Dispersion	Summarise performance, identify outliers in KPIs.	Preprocess data, standardize features for ML pipelines.
Hypothesis Testing	Validate product metrics, design & evaluate A/B tests.	Build statistical significance checks into model training.
Non-Parametric Methods	Analyze highly skewed business metrics (e.g., revenue).	Model distributions that violate normality assumptions.
Causal Inference	Evaluate historical trends and campaign effectiveness.	Model policy changes, simulate counterfactual scenarios.
Advanced Modeling	Mainly regression and trend lines for forecasting.	Bayesian networks, time-series forecasting, complex ML architectures.

Recommended 8-Week Study Sequence

Step 1 (Week 1-2): Descriptive Statistics & Spatial

Master mean, median, mode, variance, standard deviation, and standardisation via Z-scores. Code these from scratch in Python to internalize the logic.

Step 2 (Week 3): Probability & Distributions

Study normal, log-normal, binomial, and Pareto distributions. Understand how the CLT guarantees normality of sample means.

Step 3 (Week 4-5): Hypothesis & Non-Parametric Tests

Formulate null hypotheses. Master Z-tests, T-tests, and Chi-Square, then learn when to swap them for Mann-Whitney U and Kruskal-Wallis.

Step 4 (Week 6-7): Specialized Experimentation & Modeling

Analysts: Master A/B testing design, power analysis, and Bonferroni corrections.

Scientists: Dive into Bayesian probability, class imbalance adjustments (precision/recall), and causal analysis.

Step 5 (Week 8): Practical Application

Synthesize your knowledge by analyzing an open-source dataset, cleaning outliers, correcting class imbalances, and drafting a formal A/B test proposal.